AWS Lab: Private Subnet Route Table & NAT Gateway Egress Triage

Duration : 30 Min.

Scenario

EC2 instances in a private application subnet (subnet-priv-01) cannot download security patches from public package repositories. An engineer suspects a networking misconfiguration. The route table associated with the private subnet is missing a default route (0.0.0.0/0) pointing to the NAT Gateway (nat-0a1b2c3d4e5f) located in the public subnet. The engineer must inspect the route table and add the missing default egress route.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon VPC Route Tables & NAT Gateways - **Subnet:** `subnet-priv-01` (Private subnet) - **Route Table ID:** `rtb-01234` - **NAT Gateway ID:** `nat-0a1b2c3d4e5f` (Public subnet)

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Ingest VPC route table configuration from /opt/aws_net/route_table.json. 2. Diagnose why instances cannot reach external IP addresses (missing 0.0.0.0/0 destination). 3. Write an AWS CLI command to create the route: aws ec2 create-route --route-table-id rtb-01234 --destination-cidr-block 0.0.0.0/0 --nat-gateway-id nat-0a1b2c3d4e5f. 4. Output verification steps to ~/nat_route_fix.sh.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Route table inspection: parse route table entries and identify missing internet destination	3 Marks
TC2	NAT target identification: locate the operational NAT Gateway ID in the public subnet	2 Marks
TC3	Command creation: nat_route_fix.sh contains valid create-route CLI syntax	3 Marks
TC4	Verification check: script includes route table describe command to verify active status	2 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.